# Design & develop a system for disease prediction and suggest corrective actions in pigeon pea crops using AI

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*Abstract*—**The pigeon pea (Cajanus cajan) is a critical pulse crop in India and other semi-arid regions for securing food supplies as well as ensuring the soil remains healthy and continues to produce crops over time. However, with diseases such as leaf spot, sterility mosaic, and leaf webber affecting the crops, the ability to produce these high-yielding crops has been significantly hindered due to the lengthy time it takes to identify diseases and lack of access to trained professionals on how to manage them effectively. Although deep learning technology has demonstrated the capability to detect diseases in plants effectively, most of the deep learning models currently being used have been trained on generic image datasets created in controlled environments and may not work effectively in the field for site-specific diseases.**

**This research seeks to explore the use of transfer learning convolutional neural networks as a means of classifying the various diseases affecting pigeon pea plants using images captured from their natural habitat or environment. This research will examine the results of several different pre-trained architectures developed with deep learning as it relates to diseases affecting this crop type, while assessing the relative performance of each model using standardized metrics, including accuracy, precision, recall, and F1-score, as well as evaluating class-by-class metrics to determine how well the model generalizes and performs under various stressors.**

**this study aims to develop an accurate and consistent baseline for identifying pigeon pea diseases from their images, as well as demonstrating the importance of moving from multiclass/generalized deep learning technology to crop-specific, real-world agricultural deep learning products.**

***Keywords—Pigeon pea; Cajanus cajan; Plant disease classification; Deep learning; Transfer learning; Convolutional neural networks; Crop-specific modeling; Leaf image analysis; Field-based dataset; Agricultural artificial intelligence***

## I. INTRODUCTION

pigeon pea (Cajanus cajan) is an important pulse crop that is grown throughout the world, especially in tropical and semi-arid areas. Pigeon peas are an important part of the diet because they are high in proteins and Nitrogen contributes to the health of the soil by fixing it. This crop is highly susceptible to a number of plant diseases, including leaf spot, sterility mosaic and leaf Webber infestations, which have the potential to drastically affect yield and quality of pigeon pea. When those diseases are not identified early on, they can cause major losses to the producer.

Historically, disease identification in traditional agriculture has been based on using visual inspection by farmers or agricultural specialists. Because of this subjective method, it is easy to misclassify crops based on their appearance or lesions and can occur early in the Infection process before any other signs have been observed. Artificial Intelligence has created a powerful tool for plant disease detection by creating a system of deep-learning based image analysis methods. Convolutional Neural Networks (CNNs) have been used successfully at detecting visual diseases from leaf images for many different species of crops. Although the vast majority of literature examining existing plant disease datasets is focused on controlled laboratory samples, there is little attention given to real-world variability due to lack of consistency in lighting, background variables (such as noise), or the existence of overlapping leaves at a different growth stage. This makes it difficult to predict the generalizability of any model trained on these datasets when tested in a field environment. In addition, there are limited studies that specifically examine the pigeon pea crop, creating a significant gap in research on this particular crop.

Therefore, the purpose of this study is to develop transfer learning based deep learning algorithms for classifying pigeon pea leaf diseases from images created by field based datasets. The study aims to establish a reliable baseline for crop specific disease prediction across the spectrum of variability present in the agricultural environment.

| Disease Name | Type of Pathogen | Visual Symptoms | Impact on Yield |
|---|---|---|---|
| Leaf Spot | Fungal | Brown circular spots on leaves | Moderate–High |
| Sterility Mosaic | Viral | Mosaic pattern, stunted growth | High |
| Leaf Webber | Insect Pest | Web formation, leaf damage | Moderate |
| Healthy Condition | — | Uniform green leaves | — |

Table 01: Major Diseases of Pigeon Pea and Their Key Visual Symptoms

## II. Related Work

The use of deep learning in the detection of plant diseases has fundamentally changed the way plant disease research is conducted in the past decade. One of the methods that have been used to classify images of leaves is through the use of convolutional neural networks (CNN). It has been shown that CNNs have the potential to learn to extract relevant visual features from images and, therefore, work well for the classification of image data. For example, various researchers have reported very high levels of accuracy when recognizing plant diseases that use image classification networks such as ResNet, VGG, and MobileNet to analyze publically available datasets of plant diseases.

Many of the earlier studies used datasets that were collected in laboratory settings with controlled conditions (i.e, where the leaves were photographed on a constant similar-coloured background). These studies demonstrated high levels of classification performance but did not provide any performance data regarding how well these types of image classification networks would perform in real agricultural settings. The agricultural field presents challenges with uncertainty associated with light-source variation, background clutter, occlusions, and leaf positioning which have a negative impact on generalization.

Transfer learning strategies have been examined as means for addressing reduced dataset constraints in recent studies. Through the use of fine-tuning on a variety of pre-trained deep learning networks or models, accuracy classifying crops with little more than a modest amount of data was achieved; however, most of the literature surrounding this topic is limited to commonly studied crops such as potato, maize, and tomato. Additionally, the number of pigeon pea crop specific studies is lesser than with other crops.

Although CNN-based methods have been widely studied since their introduction; recent research has included both hybrid and ensemble learning methods as a way to improve the performance of detecting diseased plants. Researchers have studied the use of feature extraction methods with different classifier types (such as support vector machines and gradient boosting) to make the classifications more robust. Many of the hybrid models exhibit similar accuracy when compared to traditional models; however they are often much more computationally expensive to train and require careful selection of the ideal model parameters to use. Many of the methods evaluated were developed on a limited or homogenous dataset, meaning that their generalization ability across different environmental conditions will likely be limited. This serves as a basis for the need to evaluate transfer learning architectures in a systematic manner to develop models that can balance system performance, computational efficiency, and adaptability in crop-specific disease classification tasks such as pigeon pea.

Additionally, with regard to the use of IoT for disease monitoring and/or detection, whilst IoT is encouraging and has shown promise, their vast computational needs and infrastructure dependencies make them impractical for low-resource rural areas.

The available literature supports that deep learning approaches may be effective for plant disease classification; however, they also lack in the following areas:

1) Specificity to crop of focus.

2) validation of findings under realistic field conditions

3) systematic comparison of results with other crop species' results, particularly pigeon pea.

| **Study Focus** | **Method Used** | **Dataset Type** | **Limitation** |
|---|---|---|---|
| Generic plant disease detection | CNN-based models | Controlled lab dataset | Poor field generalization |
| Transfer learning approaches | Pre-trained ResNet/MobileNet | Mixed crop datasets | Limited crop-specific focus |
| IoT-based monitoring systems | DL + Sensor integration | Experimental setups | High system complexity |
| Lightweight models | Optimized CNN | Small datasets | Limited robustness testing |

Table 02: Comparative Summary of Existing Studies on Plant Disease Detection

## III. Research Gap and Problem Statement

Before In recent years, there has been a notable improvement in the area of plant disease detection utilizing deep learning; however, there are still limitations when applying these methods to crop-specific conditions such as those found with pigeon peas. Many current studies are based on commonly grown crops (including tomatoes, potatoes, or corn) and because of the limited amount of research conducted regarding pigeon peas (which are also significant crops), there is no existing systematic evaluation framework that is specific to classifying diseases in pigeon peas.

The majority of models that have been published were developed using controlled laboratory datasets that are consistent in background and contain very little environmental variability. These models often report high levels of accuracy when classifying disease; however, the ability of these models to perform well in a real-world scenario remains unclear. This is due to the presence of significant variations in light levels, leaf orientation, clutter on the background, and overlapping leaves. There is little or no record of these types of problems being addressed through the use of controlled datasets, and therefore, models that have been trained under ideal conditions may not generalize well when used in actual farming conditions.

The literature also cites that another significant limitation is that there is not a sufficient comparative analysis of transfer learning architecture models for crop-specific adaptation.

While CNNs that are pretrained are commonly used, there are not many references to architecture models that are best suited for Pigeon Pea disease detection in relation to field variability.

Thus, evaluating the transfer learning-based deep learning model for pigeon pea disease classification in a structured crop-specific manner using field representative datasets is warranted. The aim of this research is to determine if a carefully modified and compared deep learning framework can provide a reliable and robust prediction of disease for pigeon pea in the natural environment of agriculture.

## IV. METHODOLOGY

After This study made use of a systematic experimental methodology to evaluate the efficacy of transfer learning based deep learning models for the classification of pigeon pea leaf diseases under field conditions. The study methodology includes datasets, model selection, training strategies and the evaluation of models' performance.

### 1. Dataset Description

Pigeon pea leaf images were used as the basis of the datasets for this study. Various pigeon pea leaf diseases and healthy pigeon pea leaf samples were represented by these images. All images included in the datasets were collected in natural environmental conditions and therefore contained significant variability; including different illumination levels, varying degrees of background complexity, and differences in the growth stages of leaves. Each of these sources of variability was included in the dataset in an effort to simulate typical agricultural environments and to test the robustness of the models against those environmental conditions as opposed to testing them in a controlled laboratory setting.

The datasets used in this study were divided into training, validation, and test sets with a focus on ensuring that the datasets were balanced as much as possible. All datasets were preprocessed using standard methods (i.e., resizing and normalizing) to ensure consistency between models in terms of input to the models.

### 2. Model Selection Strategy

The intent of this study was to examine the adequacy of transfer learning based deep learning model selection for crop specific disease classification. In order to accomplish this, a number of different architectures for pre-trained convolutional neural networks (CNNs) were selected for the comparison of performance. These pre-trained architectures were trained prior to being selected for this study on extremely large image datasets and were then fine-tuned using images of pigeon pea leaves in order to adapt the features for classification to the pigeon pea crop. Authors and Affiliations.

Only adjust the high-level layers to learn features that are specific to a disease, while retaining the low-level features extracted by previously trained models; learn from previously learned visual patterns in the lower level of the models. Thus, moderately sized datasets can learn effectively.

### 3. Training/ Validation methods

We are employing a supervised learning paradigm, with leaf images being classified into their appropriate disease categories by learning the patterns of leaves through visual features. To ensure reliability and reduce overfitting, cross-validation methods will be used when training the models. For consistency between the models, hyperparameters are systematically picked.

### 4. Metrics for assessing model performance

Model performance will be evaluated based on standard classification metrics: accuracy, precision, recall, and F1-score. Overall, performance will be measured based on accuracy. However, each disease class output will also be analyzed for its performance, i.e., the ability of the model to tell two different diseases apart when presented with visually similar classes. The analysis of the confusion matrix will allow us to determine the misclassifications made by the models and help us evaluate the robustness of the model as it operates in variable conditions in the field.

This methodological design enables systematic comparison of transfer learning architectures and provides a reliable basis for assessing crop-specific disease prediction performance in pigeon pea.

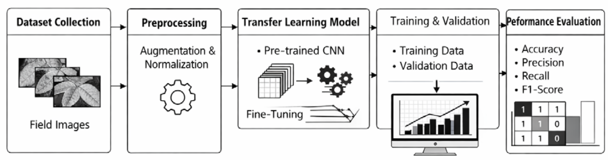


Fig_01: Methodological Framework Pigeon Pea Disease Classification

## V. EXPECTED RESULTS AND OUTCOMES

The goal of this research will be to perform a thorough comparative evaluation of existing transfer-learning based deep learning techniques which can classify diseases on pigeon pea leaves in field-like conditions. Previous research and initial observations have suggested that fine-tuned pre-trained architectures should produce better adaptation to feature extraction than those that were trained from scratch.

The anticipated outcome is that all the chosen transfer learning-based models will produce a high degree of overall classification accuracy with balanced precision and recall measures across the various disease categories, with possible higher classification error rates among those categories most like one another. Therefore, the use of class-based analysis may provide a more detailed understanding of the robustness and discriminatory performance of the models when classifying multiple classes of pigeons.

Additionally, we will compare the various architectures of the models against each other to identify differences in generalization capabilities, computational efficiency, and stability of classification. By completing these comparative evaluations, we aim to establish a valid baseline for pigeon pea disease detection in a non-laboratory setting, using representative images of the same region in which we will conduct the disease detection.

Overall, the expected outcome of this study is the identification of a suitable transfer learning framework capable of robust disease prediction under natural agricultural variability, thereby contributing to the advancement of crop-specific artificial intelligence applications in precision agriculture.

## VI. Conclusion

This research will report on the use of deep learning through transfer learning methods to provide disease classification by crop for pigeon pea. Tests have been performed for plant disease with numerous models, yet most of these models are either general dataset or laboratory controlled and may not give an accurate representation of field crops as they exist today. By limiting this research to pigeon pea, testing different pre-trained architectures for their ability to identify diseases at various levels of development or maturity, this research will provide a foundation for future development of a standardized, comparative, and clearly organized evaluation method.

This work will reflect the importance of a thorough class wise analysis (e.g., performance) of the different models evaluated; generalization ability (how well the models perform beyond the data on which they were trained); and reliability (how well they will operate in comparison to real-world environments). This study illustrates the need to develop a systematic method for comparing various models in a means that assesses both systematic method for comparing and providing analytical means for describing which methods will be the best suited for specific crops.

The anticipated outcome of this research will be the establishment of a readily available reference point for identifying pigeon pea diseases and the reinforcement of the use of artificial intelligence for use in precision agriculture. Future work may enhance the use of this framework, incorporating additional environmental parameters; utilizing larger multi-seasonal sets of data; and incorporating the ability of the system to act on real-time data to improve scalability and applicability of the systems in-field.

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